

Data Mining Course: Conda Environment Setup Instructions

In preparation for our upcoming exercise for the Data Mining course, please ensure that your laptops are fully set up to work with Python and that you bring the laptops to our sessions.

Follow these steps according to your operating system to set up Python, Conda, and essential libraries for this module.

1. Setting up Python on Your Operating System:

1.1. Windows Users: Install Windows Subsystem for Linux (WSL)

WSL allows you to run a Linux terminal on Windows, providing an easier environment for Python and data science tools. Here's you can find how to install and set it on your laptop: [\[How to install Linux on Windows with WSL\]](#)

1.2. macOS & Linux Users: Use Terminal

For Mac and Linux users, no additional software is needed; the terminal is built into your operating system.

2. Installing Miniconda (alternatively use Anaconda):

Anaconda is a package manager, environment manager, and distribution of Python, with many data science libraries included. It simplifies the installation and management of Python packages and their dependencies. To download and install Conda:

1. go to [\[Miniconda installation page, Anaconda Installation page\]](#) and download the installer for your operating system.
2. Install Miniconda/Anaconda: Run the installer and follow the instructions. Follow the prompts to complete the installation.
3. Add Codna to PATH.
4. During installation, you may be prompted to add Conda to PATH. This allows you to call `conda` from any terminal window.

3. Installing Required Python Libraries:

To run machine learning and data science code effectively, install the following libraries:(NumPy, Pandas, Matplotlib & Seaborn, SciKit-Learn, Jupyter , PyTorch & TensorFlow).

To install these libraries, you can use the Yaml file I exported from an environment I created as described below with all packages needed in this course:

```
```bash:
conda env create -f environment.yml
```
```

This instruction will create an environment named DataMining and install all packages listed in the Yaml file.

Alternatively (this is how I created the environment): open the Anaconda Prompt or terminal, then create a virtual environment (recommended for isolating dependencies):

```
```bash:
conda create -n DataMining python=3.9
conda activate DataMining
```
```

Now you are working in your Data Mining environment. Next, install the packages:

```
```bash
conda install numpy pandas matplotlib seaborn scikit-learn jupyter conda
install pytorch torchvision torchaudio -c pytorch
conda install tensorflow
```
```

4. Testing the Installation

After setting up everything, verify your installation by opening a Notebook. To do so:

1. Open the terminal or Anaconda Prompt. Make sure you have activated conda environment DataMining by typing.

```
```bash
conda activate DataMining
```
```

2. Start Jupyter by typing:

```
```bash
jupyter notebook .
```
```

3. In the notebook interface, try importing the libraries to confirm installation:

```
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
import torch
import tensorflow as tf
```
```

5. Explanation of Key Components:

- NumPy: Provides support for large, multi-dimensional arrays and matrices, along with mathematical functions.
- Pandas: Used for data manipulation and analysis, with data structures and operations for handling datasets.
- Matplotlib: A plotting library that helps visualize data with charts and graphs.
- Seaborn: Based on Matplotlib, used for statistical data visualization and aesthetic data representation.
- SciKit-Learn: A comprehensive machine learning library with tools for model training, evaluation, and data preprocessing.
- Jupyter: An interactive notebook that allows running code, visualizations, and documentation in one place.
- PyTorch: An open-source deep learning framework developed by Facebook, widely used for neural networks.
- TensorFlow: An end-to-end open-source machine learning platform developed by Google, known for deep learning.

6. Additional Tips

- Stay Updated: Periodically update Conda and libraries using ``conda update -all`` to ensure you have the latest features and security patches.
- Resource Management: Close Jupyter Notebooks when not in use to free up memory, especially if working with large datasets (everything you load in the notebook will end up in the memory)
- Practice: Familiarize yourself with each tool by practicing small projects or tutorials to gain confidence in using them. The internet is full of resources that you could use to learn python and data science.